Puram 2017 HNSCC — recapitulation summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → PASS

10/17 samples (≥100 cells) intermix on one continuous manifold; no per-patient islands (64 detached cells removed).

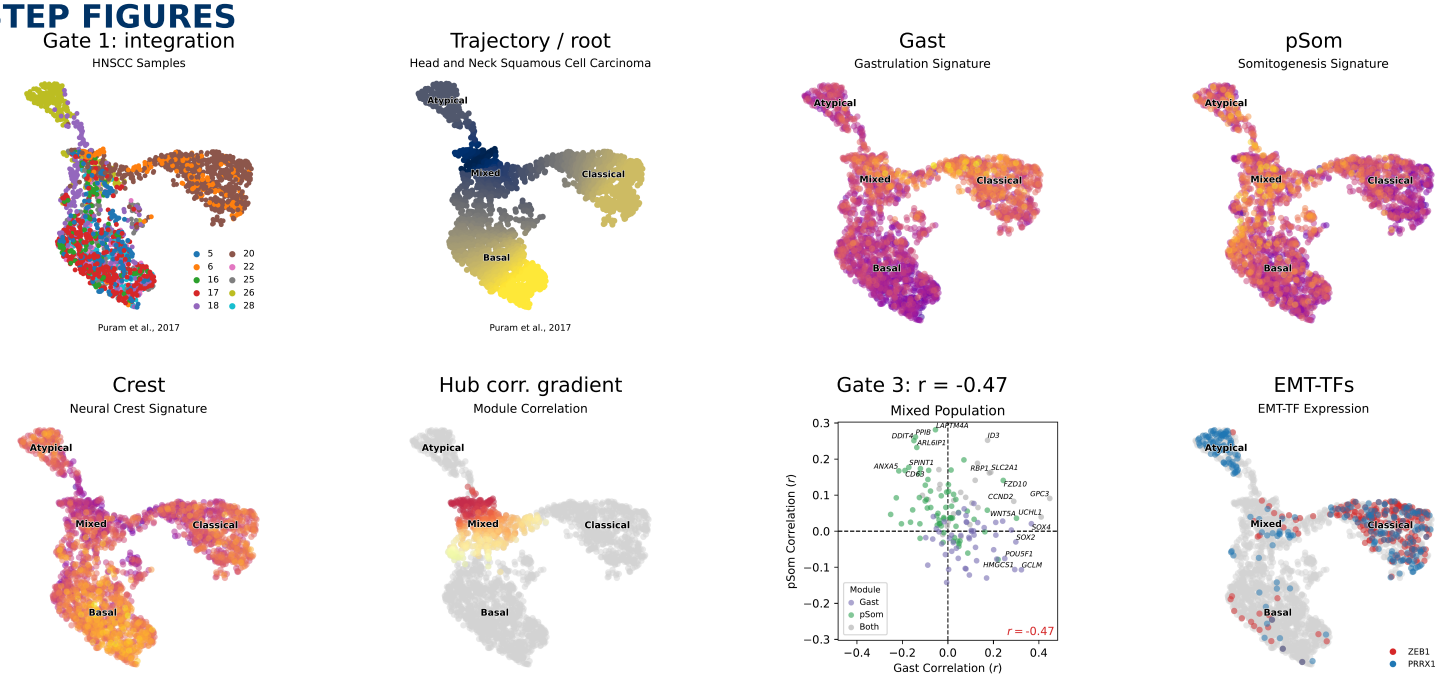
Gate 2 — module geography → PASS

Gast/pSom hub diverges into arms; Crest is a terminal readout at the Basal tip (0.43 > Atypical 0.27 ≈ Classical 0.24).

Gate 3 — hub mutual exclusivity → PASS

On 466 Mixed-hub cells, per-gene Gast vs pSom correlation vectors anti-correlate: r = -0.47 (ref ≈ -0.49).

STEP FIGURES



Dataset. Head & Neck SCC (Puram et al., 2017), SmartSeq2 — 18 patients / 5,902 cells (master scope); malignant only: 2,539 load, 2,408 across 4 subtypes enter the trajectory.

Verdict. ✓ Pattern reproduces — all three gates pass. Gast/pSom mutually exclusive within the Mixed hub (r = -0.47 vs ref -0.49), hub at the root of a hub-and-spoke manifold; EMT-TFs ZEB1 (Classical) / PRRX1 (Atypical).

Parameter choices

Min cells/sample	≥ 100 (keeps 10/17)	**Integration**	Harmony on sample, 20 it
Island removal	UMAP radius-graph (−64 cells)	**Trajectory**	scFates PPT N=200 λ=2000 σ=0.5
Root	node 85 = Mixed hub (466 cells)	**Subtypes**	markers + Puram 2017